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STENT WITH ANTI-MIGRATION FEATURES

Abstract

An illustrative stent may comprise an elongated tubular member having a first end and a second end and an intermediate region disposed therebetween. The elongated tubular member configured to move between a collapsed configuration and an expanded configuration. The elongated tubular member may comprise at least one twisted filament, such as a knitted filament having a plurality of twisted knit stitches with intermediate rung portions extending between adjacent twisted knit stitches and one or more anti-migration features formed in one or more intermediate rung portions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/487,937, filed Sep. 28, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 63/084,865 filed on Sep. 29, 2020, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure pertains to medical devices, methods for manufacturing medical devices, and uses thereof. More particularly, the present disclosure pertains to a stent for implantation in a body lumen, and associated methods.

BACKGROUND

[0003] A wide variety of intracorporeal medical devices have been developed for medical use, for example, surgical and/or intravascular use. Some of these devices include guidewires, catheters, medical device delivery systems (e.g., for stents, grafts, replacement valves, etc.), and the like. These devices are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods.

SUMMARY

[0004] This disclosure provides design, material, manufacturing method, and use alternatives for medical devices. An example medical device may include a stent.

[0005] In a first example, a stent may comprise an elongated tubular member comprising at least one knitted filament having a plurality of twisted knit stitches with intermediate rung portions extending between adjacent twisted knit stitches. The elongated tubular member may be configured to move between a collapsed configuration and an expanded configuration. The elongated tubular member may further comprise one or more anti-migration features formed in one or more of the intermediate rung portions. When the elongated tubular member is in the expanded configuration, the one or more anti-migration features may extend radially therefrom.

[0006] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may extend in the range of 1 to 4 millimeters radially beyond a base diameter of the elongated tubular member in the expanded configuration.

[0007] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may extend at a non-parallel angle relative to a longitudinal axis of the elongated tubular body.

[0008] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may include a distally oriented apex.

[0009] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may bend back on itself.

[0010] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may each include a first joint bend adjacent a first twisted knit stitch and a second joint bend adjacent a second twisted knit stitch.

[0011] Alternatively or additionally to any of the examples above, in another example, the first and second joint bends may be configured to cause the anti-migration loop to lie flat as the elongated tubular body is moved from the expanded configuration to the collapsed configuration.

[0012] Alternatively or additionally to any of the examples above, in another example, when the

elongated tubular member is in the collapsed configuration at least a portion of the one or more anti-migration features may be subsumed into one or more adjacent twisted knit stitches.

[0013] Alternatively or additionally to any of the examples above, in another example, a length of the intermediate rung portions in the collapsed configuration may be less than a length of the intermediate rung portions in the expanded configuration.

[0014] Alternatively or additionally to any of the examples above, in another example, at least some of the plurality of twisted knit stitches are suspended from a twisted knit stitch in a preceding row.

[0015] Alternatively or additionally to any of the examples above, in another example, a diameter of the elongated tubular member in the collapsed configuration may be in the range of about 60% to 80% less than a diameter of the elongated tubular member in the expanded configuration.

[0016] In another example, a method of manufacturing a stent having anti-migration features may comprise knitting a stent blank, the knitted stent blank may include at least one knitted filament having a plurality of twisted knit stitches with intermediate rung portions extending between adjacent twisted knit stitches, disposing a knitted stent blank in position over a mandrel, the mandrel including one or more anti-migration feature forming elements, engaging one or more intermediate rung portions of the knitted stent with the one or more anti-migration feature forming elements to form an anti-migration feature having a first point bend and a second point bend, annealing the woven stent blank while disposed on the mandrel to form a shaped stent with the anti-migration feature, and disengaging the one or more anti-migration feature forming elements in order to remove the shaped stent from the mandrel.

[0017] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration feature forming elements may comprise pins that are configured to be driven in a radially outward direction relative to a central longitudinal axis of the mandrel, and engaging the wire with the one or more anti-migration feature forming elements may comprise driving the pins in the radially outward direction relative to the central longitudinal axis of the mandrel.

[0018] Alternatively or additionally to any of the examples above, in another example, disengaging the one or more anti-migration feature forming elements may comprise permitting the pins to move in a radially inward direction relative to the central longitudinal axis of the mandrel.

[0019] Alternatively or additionally to any of the examples above, in another example, disposing the knitted stent blank in position over the mandrel may comprise stretching the knitted stent blank over the mandrel and allowing the knitted stent blank to conform to an outer surface of the mandrel.

[0020] In another example, a stent may comprise an elongated tubular member comprising at least one knitted filament having a plurality of twisted knit stitches with intermediate rung portions extending between adjacent twisted knit stitches, the elongated tubular member may be configured to move between a collapsed configuration and an expanded configuration and one or more anti-migration features formed in one or more of the intermediate rung portions. When the elongated tubular member is in the expanded configuration, the one or more anti-migration features may extend radially therefrom.

[0021] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may extend in the range of 1 to 4 millimeters radially beyond a base diameter of the elongated tubular member in the expanded configuration.

[0022] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may extend at a non-parallel angle relative to a longitudinal axis of the elongated tubular body.

[0023] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may include a distally oriented apex.

[0024] Alternatively or additionally to any of the examples above, in another example, when the

elongated tubular member is in the expanded configuration the one or more anti-migration features may bend back on itself.

[0025] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may each includes a first joint bend adjacent a first twisted knit stitch and a second joint bend adjacent a second twisted knit stitch.

[0026] Alternatively or additionally to any of the examples above, in another example, the first and second joint bends may be configured to cause the anti-migration loop to lie flat as the elongated tubular body is moved from the expanded configuration to the collapsed configuration.

[0027] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the collapsed configuration at least a portion of the one or more anti-migration features may be subsumed into one or more adjacent twisted knit stitches.

[0028] Alternatively or additionally to any of the examples above, in another example, a length of the intermediate rung portions in the collapsed configuration may be less than a length of the intermediate rung portions in the expanded configuration.

[0029] Alternatively or additionally to any of the examples above, in another example, a diameter of the elongated tubular member in the collapsed configuration may be in the range of about 60% to 80% less than a diameter of the elongated tubular member in the expanded configuration.

[0030] In another example, a stent may comprise an elongated tubular member comprising at least one knitted filament having a plurality of twisted knit stitches each including a loop portion and an overlapping base region with intermediate rung portions extending between adjacent twisted knit stitches and at least some of the plurality of twisted knit stitches are suspended from an intermediate rung portion of a preceding row, the elongated tubular member configured to move between a collapsed configuration and an expanded configuration and one or more anti-migration features formed in one or more of the intermediate rung portions. When the elongated tubular member is in the expanded configuration, the one or more anti-migration features may extend radially therefrom.

[0031] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may be positioned at a similar longitudinal location about a circumference of the elongated tubular member.

[0032] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may extend in the range of 1 to 4 millimeters radially beyond a base diameter of the elongated tubular member in the expanded configuration.

[0033] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may extend at a non-parallel angle relative to a longitudinal axis of the elongated tubular body.

[0034] Alternatively or additionally to any of the examples above, in another example, when the elongated tubular member is in the expanded configuration the one or more anti-migration features may bend back on itself.

[0035] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration features may each include a first joint bend adjacent an overlapping base region of a first twisted knit stitch and a second joint bend adjacent an overlapping base region of a second twisted knit stitch.

[0036] In another example, a method of manufacturing a stent having anti-migration features may comprise knitting a stent blank, the knitted stent blank including at least one knitted filament having a plurality of twisted knit stitches with intermediate rung portions extending between adjacent twisted knit stitches, disposing a knitted stent blank in position over a mandrel, the mandrel including one or more anti-migration feature forming elements, engaging one or more intermediate rung portions of the knitted stent with the one or more anti-migration feature forming elements to form an anti-migration feature having a first point bend and a second point bend, annealing the woven stent blank while disposed on the mandrel to form a shaped stent with the

anti-migration feature, and disengaging the one or more anti-migration feature forming elements in order to remove the shaped stent from the mandrel.

[0037] Alternatively or additionally to any of the examples above, in another example, the one or more anti-migration feature forming elements may comprise pins that are configured to be driven in a radially outward direction relative to a central longitudinal axis of the mandrel, and engaging the wire with the one or more anti-migration feature forming elements comprises driving the pins in the radially outward direction relative to the central longitudinal axis of the mandrel.

[0038] Alternatively or additionally to any of the examples above, in another example, disengaging the one or more anti-migration feature forming elements may comprise permitting the pins to move in a radially inward direction relative to the central longitudinal axis of the mandrel.

[0039] Alternatively or additionally to any of the examples above, in another example, disposing the knitted stent blank in position over the mandrel may comprise stretching the knitted stent blank over the mandrel and allowing the knitted stent blank to conform to an outer surface of the mandrel.

[0040] The above summary of some embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] The invention may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying drawings, in which:

[0042] FIG. 1 is a side view of an illustrative stent;

[0043] FIG. 2 is an enlarged side view of a portion of the illustrative stent of FIG. 1;

[0044] FIG. 3 is a partial side view of the illustrative stent of FIG. 1 in an elongated configuration;

[0045] FIG. 4 is an illustrative method of forming a stent;

[0046] FIG. 5 is a perspective view of an adjustable mandrel;

[0047] FIG. 6 is an exploded perspective view of the adjustable mandrel of FIG. 5;

[0048] FIG. 7 is a cross-sectional view of the adjustable mandrel of FIG. 5, with the anti-migration feature forming pins shown in a fully extended position;

[0049] FIG. 8 is a cross-sectional view of the adjustable mandrel of FIG. 5, with the anti-migration feature forming pins shown in a partially extended position;

[0050] FIG. 9 is a perspective view of an anti-migration feature forming pin forming a portion of the adjustable mandrel of FIG. 5;

[0051] FIG. 10 is a side view of a portion of the adjustable mandrel of FIG. 5, showing a portion of a knitted stent disposed about the adjustable mandrel;

[0052] FIG. 11A is a side view of the illustrative knitted stent of FIG. 10 removed from the adjustable mandrel;

[0053] FIG. 11B is a side view of another illustrative knitted stent removed from the adjustable mandrel;

[0054] FIG. 11C is a side view of another illustrative knitted stent removed from the adjustable mandrel;

[0055] FIGS. 12A-12D are schematic illustrations of anti-migration features that a knitted stent may include;

[0056] FIG. 13 is a partial side view of another illustrative knitted stent having one or more anti-migration features;

[0057] FIG. 14 is a side view of region A of the stent of FIG. 13;

[0058] FIG. 15 is a side view of the stent of FIG. 14 in a second configuration;
[0059] FIG. 16 is a side view of an illustrative delivery system for delivering a stent; and
[0060] FIG. 17 is a side view of the illustrative delivery system of FIG. 16 with the stent in a partially deployed configuration.

[0061] While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the scope of the invention.

DETAILED DESCRIPTION

[0062] For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

[0063] All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In many instances, the term “about” may be indicative as including numbers that are rounded to the nearest significant figure.

[0064] The recitation of numerical ranges by endpoints includes all numbers within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

[0065] Although some suitable dimensions ranges and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges and/or values may deviate from those expressly disclosed.

[0066] As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

[0067] The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The detailed description and the drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention. The illustrative embodiments depicted are intended only as exemplary. Selected features of any illustrative embodiment may be incorporated into an additional embodiment unless clearly stated to the contrary.

[0068] In some instances, it may be desirable to provide an endoluminal implant, or stent, that can deliver luminal patency in a patient with an esophageal stricture or other medical condition. Such stents may be used in patients experiencing dysphagia, sometimes due to esophageal cancer. An esophageal stent may allow a patient to maintain nutrition via oral intake during cancer treatment or palliation periods. Current gastrointestinal (GI) stenting regimes for the treatment of vessel strictures may rely on a self-expanding stent (SES) to resolve underlying stricture while remaining in-situ. However, this type of stenting may have a high prevalence for migration, particularly in fully covered designs. Some stents may include protrusions (e.g., loops, quills, etc.) that radially protrude out from the stent body. These raised features may interact with the duct of vessel to reduce device migration. However, these devices may be difficult to remove or reposition as they may be difficult to purse down to lower diameters. Further, devices with these types of protrusions (or anti-migration features) may be indicated for permanent implantation. Where these devices are desired to be removable, retraction or repositioning of the devices may be limited as the angulation of the protrusions may cause vessel damage with movement of the stent. What may be desirable is an endoluminal implant or stent that includes anti-migration features and can also be readily pursed down for repositioning and/or removal. While the embodiments disclosed herein are discussed with reference to esophageal stents, it is contemplated that the stents described herein may be used and

sized for use in other locations such as, but not limited to: bodily tissue, bodily organs, vascular lumens, non-vascular lumens and combinations thereof, such as, but not limited to, in the coronary or peripheral vasculature, trachea, bronchi, colon, small intestine, biliary tract, urinary tract, prostate, brain, stomach and the like.

[0069] FIG. 1 illustrates a side view of an illustrative endoluminal implant **10**, such as, but not limited to, a stent. In some instances, the stent **10** may be formed from an elongated tubular member **12**. While the stent **10** is described as generally tubular, it is contemplated that the stent **10** may take any cross-sectional shape desired. The stent **10** may have a first, or proximal end **14**, a second, or distal end **16**, and an intermediate region **18** disposed between the first end **14** and the second end **16**. The stent **10** may include a lumen **20** extending from a first opening adjacent the first end **14** to a second opening adjacent to the second end **16** to allow for the passage of food, fluids, etc.

[0070] The stent **10** may be expandable from a first radially collapsed configuration (not explicitly shown) to a second radially expanded configuration. In some cases, the stent **10** may be deployed to a configuration between the collapsed configuration and a fully expanded configuration. The stent **10** may be structured to extend across a stricture and to apply a radially outward pressure to the stricture in a lumen to open the lumen and allow for the passage of foods, fluids, air, etc. When the stent **10** is in a radially collapsed configuration, the outer diameter of the stent **10** is reduced relative to a fully or partially expanded configuration. In some cases, to reduce the diameter of the stent **10** for delivery to the target location, the stent **10** is stretched or lengthened. When the stent **10** is deployed (moved from the delivery configuration to an expanded configuration), the length of the stent **10** decreases and the diameter increases. In some cases, stents **10** may experience in the range of about 20% to about 40% foreshortening (e.g., the percentage by which the length of a stent decreases from its delivery configuration to its expanded configuration). It is contemplated that the change in length and/or the change in diameter of the stent **10** may be at least partially dependent on a size (e.g., diameter) of the stent. In some cases, a biliary stent may have a deployed diameter in the range of about 8 millimeters to about 10 millimeters and a constrained diameter of about 2.5 millimeters to about 3 millimeters. This may correspond to about a 60% to about 80% reduction in diameter from the expanded configuration to the delivery configuration. This is just an example. The reduction in diameter may be less than 60% or greater than 80%, as desired. In another example, an endoscopy stent may have a deployed diameter in the range of about 18 millimeters to about 23 millimeters and a constrained diameter of about 6 millimeters to about 6.5 millimeters. This may correspond to about a 60% to about 80% reduction in diameter from the expanded configuration to the delivery configuration. This is just an example. The reduction in diameter may be less than 60% or greater than 80%, as desired.

[0071] The proximal end **14** of the stent **10** may include a plurality of loops **22**. The loops **22** may be configured to receive a retrieval tether or suture interwoven therethrough, or otherwise passing through one or more of the loops **22**. The retrieval suture may be used to collapse and retrieve the stent **10**, if so desired. For example, the retrieval suture may be pulled like a drawstring to radially collapse the proximal end **14** of the stent **10** to facilitate removal of the stent **10** from a body lumen.

[0072] The stent **10** may have a knitted structure, fabricated from a single filament **24** interwoven with itself and defining open cells **25**. In some cases, the filament **24** may be a monofilament, while in other cases the filament **24** may be two or more filaments wound, braided, or woven together. In some instances, an inner and/or outer surface of the stent **10** may be entirely, substantially, or partially, covered with a polymeric covering or coating. The covering or coating may extend across and/or occlude one or more, or a plurality of the cells **25** defined by the struts or filaments **24**. The covering or coating may help reduce food impaction and/or tumor or tissue ingrowth.

[0073] It is contemplated that the stent **10** can be made from a number of different materials such as, but not limited to, metals, metal alloys, shape memory alloys and/or polymers, as desired, enabling the stent **10** to be expanded into shape when accurately positioned within the body. In

some instances, the material may be selected to enable the stent **10** to be removed with relative ease as well. For example, the stent **10** can be formed from alloys such as, but not limited to, Nitinol and Elgiloy®. Depending on the material selected for construction, the stent **10** may be self-expanding (i.e., configured to automatically radially expand when unconstrained). In some embodiments, fibers may be used to make the stent **10**, which may be composite fibers, for example, having an outer shell made of Nitinol having a platinum core. It is further contemplated the stent **10** may be formed from polymers including, but not limited to, polyethylene terephthalate (PET). In some embodiments, the stent **10** may be self-expanding while in other embodiments, the stent **10** may be expanded by an expansion device (such as, but not limited to a balloon inserted within the lumen **20** of the stent **10**). As used herein the term “self-expanding” refers to the tendency of the stent to return to a preprogrammed diameter when unrestrained from an external biasing force (for example, but not limited to, a delivery catheter or sheath). The stent **10** may include a one-way valve, such as an elastomeric slit valve or duck bill valve, positioned within the lumen **20** thereof to prevent retrograde flow of gastrointestinal fluids.

[0074] In some instances, in the radially expanded configuration, the stent **10** may include a first end region **26** proximate the proximal end **14** and a second end region **28** proximate the second end **16**. In some embodiments, the first end region **26** and the second end region **28** may include retention features or anti-migration flared regions (not explicitly shown) having enlarged diameters relative to the intermediate portion **18**. The anti-migration flared regions, which may be positioned adjacent to the first end **14** and the second end **16** of the stent **10**, may be configured to engage an interior portion of the walls of the esophagus or other body lumen. In some embodiments, the retention features, or flared regions may have a larger diameter than the cylindrical intermediate region **18** of the stent **10** to prevent the stent **10** from migrating once placed in the esophagus or other body lumen. It is contemplated that a transition from the cross-sectional area of the intermediate region **18** to the retention features or flared regions may be gradual, sloped, or occur in an abrupt step-wise manner, as desired.

[0075] In some embodiments, the first anti-migration flared region may have a first outer diameter and the second anti-migration flared region may have a second outer diameter. In some instances, the first and second outer diameters may be approximately the same, while in other instances, the first and second outer diameters may be different. In some embodiments, the stent **10** may include only one or none of the anti-migration flared regions. For example, the first end region **26** may include an anti-migration flare while the second end region **28** may have an outer diameter similar to the intermediate region **18**. It is further contemplated that the second end region **28** may include an anti-migration flare while the first end region **26** may have an outer diameter similar to an outer diameter of the intermediate region **18**. In some embodiments, the stent **10** may have a uniform outer diameter from the first end **14** to the second end **16**. In some embodiments, the outer diameter of the intermediate region **18** may be in the range of 15 to 25 millimeters. The outer diameter of the anti-migration flares may be in the range of 20 to 30 millimeters. It is contemplated that the outer diameter of the stent **10** may be varied to suit the desired application.

[0076] The stent **10** may further include one or more radially extending anti-migration features **30**. As will be described in more detail herein, the anti-migration features **30** may be portions or loops of the filament **24** which extend radially from the stent body **12**. For example, the anti-migration features **30** may have an outer diameter that is greater than the intermediate region **18** and/or greater than an anti-migration flared region (if so provided). In some cases, the anti-migration features **30** may extend in the range of about 1 millimeter to about 4 millimeters beyond the base diameter of the stent **10**. The base diameter may be the nominal (e.g., substantially constant) outer diameter of the body **12** of the stent **10** taken at a same longitudinal location as the anti-migration features **30**. When the stent **10** is in the expanded configuration, the anti-migration features **30** may extend at a non-parallel angle relative to a longitudinal axis of the stent **10**. It is contemplated that an angle of the anti-migration features **30** may be determined, at least in part, on a length of the

anti-migration features **30**. For example, to reduce the risk of puncture, longer anti-migration features **30** may be oriented at smaller angles relative to the longitudinal axis of the stent **10** than shorter anti-migration features. This is just an example. In some cases, when the stent **10** is covered or partially covered with a covering or coating, the anti-migration features **30** may be left bare or uncovered (e.g., free from a covering or coating) to allow for tissue ingrowth to further reduce migration of the stent **10**. However, this is not required. In some cases, the anti-migration features **30** may include the covering or coating.

[0077] In some cases, the anti-migration features **30** may be circumferentially arranged about a perimeter of the stent **10** at a similar longitudinal position. However, it is contemplated that the anti-migration features **30** may be arranged at one or more different locations along the length of the stent **10**, if desired. It is further contemplated the anti-migration features **30** may be provided in circumferential rows or as discrete features (e.g., not necessarily extending about a circumference of the stent **10**), as desired. When provided as discrete features, the anti-migration features **30** may be asymmetrically or randomly positioned or arranged in a pattern that does not necessarily include complete circumferential rows. For example, patterns may include, but are not limited to staggered or alternating anti-migration features **30**.

[0078] In some cases, the anti-migration features **30** may be formed such that an apex **32** of the anti-migration features **30** is angled in a generally distal direction. Such an orientation may help limit distal movement of the stent **10** when implanted. However, other configurations may be used as desired. In some cases, the anti-migration features **30** may be formed such that the apex **32** is angled in a proximal direction. In yet other cases, the stent **10** may include a combination of distally angled and proximally angled anti-migration features **30**. It is contemplated that the apex **32** of the anti-migration features **30** may be curved or bent back over a length of the anti-migration features **30** such that the apex **32** is not in contact with the tissue (e.g., to form a more atraumatic anti-migration feature), as will be described in more detail herein.

[0079] It is contemplated that the stent **10** can be made from a number of different materials such as, but not limited to, metals, metal alloys, shape memory alloys and/or polymers, as desired, enabling the stent **10** to be expanded into shape when accurately positioned within the body. In some instances, the material may be selected to enable the stent **10** to be removed with relative ease as well. For example, the stent **10** can be formed from alloys such as, but not limited to, Nitinol and Elgiloy®. Depending on the material selected for construction, the stent **10** may be self-expanding or require an external force to expand the stent **10**. In some embodiments, composite filaments may be used to make the stent **10**, which may include, for example, an outer shell or cladding made of Nitinol and a core formed of platinum or other radiopaque material. It is further contemplated the stent **10** may be formed from polymers including, but not limited to, polyethylene terephthalate (PET). In some instances, the filaments of the stent **10**, or portions thereof, may be bioabsorbable or biodegradable, while in other instances the filaments of the stent **10**, or portions thereof, may be biostable.

[0080] FIG. 2 illustrates a partial enlarged side view of the knitted configuration of the stent **10**. The stent **10** may include a plurality of rows **50a**, **50b**, **50c**, **50d** (collectively, **50**) extending circumferentially about the stent **10**. The stent **10** may include any number of rows **50** desired. For example, the number of rows **50** may be selected to achieve a desired length of the stent **10**. The uppermost, or first, row **50a** may be unsecured and active. In some instances, the first row **50a** may include a plurality of loops **60a**, **60b**, **60c** (collectively, **60**). The loops **60** may each include a loop portion **62a**, **62b**, **62c** (collectively, **62**) and an overlapping base portion **64a**, **64b**, **64c** (collectively, **64**). The overlapping base portion **64a**, **64b**, **64c** is understood as the portion of the loops **60** in which one segment of the filament overlaps or crosses over a second segment of the filament, with the segment of the filament forming the loop portion **62a**, **62b**, **62c** extending therebetween. Adjacent loops **60** may be interconnected by a rung section **66a**, **66b** (collectively, **66**). For example, a first rung section **66a** may extend between the base portion **64a** of the first loop **60a** and

the base portion **64b** of the second loop **60b**. The next row **50b** may be suspended from the loops **60** of the first row **50a**. For example, the second row **50b** may include a plurality of loops **70a**, **70b**, **70c** (collectively, **70**) each including a loop portion **72a**, **72b**, **72c** (collectively, **72**) and a base portion **74a**, **74b**, **74c** (collectively, **74**). Adjacent loops **70** may be interconnected by a rung section **76a**, **76b** (collectively, **76**). As the stent **10** is knitted, the loop portion **72** may be wrapped about the base portion **64** of the preceding row **50a**.

[0081] It is contemplated that a single row **50** may be formed at a time. For example, the rows may be formed in succession with a subsequent row (e.g., row **50b**) being formed after the preceding row (e.g., row **50a**) has formed a complete rotation. While not explicitly shown, the loops **60** of the first row **50a** may be wrapped about a section of the filament **24** free from loops. As described herein, the loops **70** of the second row **50b** may be wrapped about the base portion **64** of the loops **60** the preceding row **50a**. For example, the filament **24** may be knitted such that it extends from the first rung section **76a**, is wrapped about the base portion **64b** of the preceding row **50a**, crosses back over itself to form base section **74b** and continues to the next rung section **76b**. It is contemplated that the loop portion **70** may be positioned on a first side of the rungs **66a**, **66b** and on a second opposite side of the loop portion **62b**. In other words, the filament **24** may be wound such that it extends on top of the second rung portion **66b**, behind the base portion **64b**, and over the first rung portion **66a** before crossing over itself to form the base portion **74b** of the loop **70b** of the second row **50b**. The reverse configuration is also contemplated in which the filament **24** may be wound such that it extends behind the second rung portion **66b**, over or on top of the base portion **64b**, and behind the first rung portion **66a** before crossing over itself to form the base portion **74b** of the loop **70b** of the second row **50b**.

[0082] The knitted structure of the stent **10** may allow the loop sections **62**, **72** to lengthen or contract such that the cells **25** and/or loop sections **62**, **72** have a first profile when the stent **10** is in the expanded configuration and a second profile, different from the first profile, when the stent **10** is in a collapsed delivery configuration. Lengthening of the loop sections **62**, **72** may allow the cross-sectional diameter of the stent **10** to be reduced for delivery. To lengthen, the loops **60**, **70** use some of the length of the filament **24** from the rungs **66**, **76** to elongate. FIG. 3 illustrates a portion of the stent **10** in an elongated configuration. As can be seen, as the loops **60**, **70** elongate, the rung material **66**, **76** is pulled into the loop portion **62**, **72** to allow for loop elongation (e.g., in a direction along a longitudinal axis **80**) while the intermediate rung portion **66**, **76** is shortened. The rung material **66**, **76** may be accessible and readily subsumed into the loop portion **62**, **72** due to the twist region **64**, **74**. Similarly, the anti-migration features **30** are readily subsumed into the loop portions. This may result in the stent **10** being constrained at lower forces allowing it to be loaded into a coaxial delivery system. It is contemplated that the knit structure of the stent **10** may be less subject to wire breaks due to fatigue from peristaltic motion, when compared to previous knit for stents. The softer curvature of the current knit pattern may allow the loops **60**, **70** be easily pursed by external forces which may be applied to the stent **10** by the anatomy. It is contemplated that stent **10** may be heat treated or annealed in the expanded configuration such that when the external force is released, the anti-migration features **30** may readily resume the radially outwardly extending configuration while the diameter of the stent **10** expands.

[0083] Once the stent **10** has been deployed, the stent **10** can be elongated (thus reducing the diameter) to remove or reposition the stent **10**. To collapse the stent **10** for repositioning or removal, the stent **10** may be actuated from either or both of the proximal end **14** or the distal end **16** of the stent **10**. In some cases, a pull wire may be used to facilitate collapse of the stent **10**. However, other actuation mechanisms may be used as desired. For example, a physician may use one or more forceps to grip one or both ends of the stent **10**. It is contemplated that in the range of about 60% to about 80% of the length of the stent **10** may experience a diameter reduction before the stent **10** easily moves within the body lumen. As the diameter of the stent **10** may reduce first at the end experiencing the actuation force, the clinician can select an end of the stent **10** to actuate

based on a desired movement of the stent **10**.

[0084] Returning to FIG. 2, the anti-migration features **30** may be formed in the interconnecting rung sections **96a**, **96b** (collectively, **96**) of one of the rows **50d**. While FIG. 2 illustrates the anti-migration features **30** in a single row **50d**, it is contemplated that any number of the rows **50** at any longitudinal location may include anti-migration features **30**. It is further contemplated that not every rung **96** may be formed into an anti-migration feature **30**. For example, in some cases, every other rung **96** may be formed into an anti-migration feature. Other patterned configurations, asymmetric configurations, and/or random configurations for the anti-migration features **30** may be used as desired. The anti-migration features **30** may extend radially outward from the body **12** of the stent **10** and at a generally non-parallel angle relative to the longitudinal axis **80**. In some cases, the apexes **32** of the anti-migration features **30** may be pointed in a generally distal direction. In other cases, the apexes **32** of the anti-migration features **30** may be pointed in a generally proximal direction (see, for example, FIG. 11B). In yet other cases, the stent **10** may include one or more anti-migration features **30** pointing distally and one or more anti-migration features **30** pointing proximally. It is further contemplated that one or more of the apexes **32** may be curled back on itself such that the apex **32** is generally pointed in a direction different from the general direction of the anti-migration features **30** (see, for example, FIG. 11C). In some cases, the stent **10** may be formed by first forming a constant diameter blank, then stretching the constant diameter stent blank over a mandrel prior to a shaping process and/or an annealing process. However, this is not required. In some cases, the stent **10** may be formed by knitting the stent **10** directly over a shaped mandrel. The formation of the anti-migration features **30** will be described in more detail herein.

[0085] FIG. 4 illustrates a side view of an illustrative stent **100** being formed about a constant diameter mandrel **110**. The stent **100** may be similar in form and function to the stent **10** described above. The stent **100** may be formed from a single knitted strand or filament **120**. In general, the stent **100** is formed by knitting in a single direction. For example, in the embodiments illustrated in FIG. 4, the strand **120** is knitted in a counterclockwise direction as shown at arrow **160**. However, it should be understood that the stent **100** may be formed by knitting in a clockwise direction, as desired. The strand **120** may follow a looped path about the mandrel **110** configured to form a plurality of interconnected loops.

[0086] The strand **120** may be manipulated (e.g., knitted) into a plurality of rows **130**, **132**, **134**, **136**, **138** each having a plurality of interconnected or intermeshing loops **140a-c**, **142a-c**, **144a-c**, **146a-c**, **148d-e**. The stent **100** may include as many rows as required to form a stent **100** having the desired length. As described above, the loops may be loosely knit and include interconnecting intermediate rung portions such as the rung portions **152a** and **152b** interconnecting three loops **146d**, **146c**, **146b** of one of the rows **136**. It should be understood that as the stent **100** is formed from a single strand **120**, the rows **130**, **132**, **134**, **136**, **138** may not be distinct and separate rows but instead form a continuous connection with the preceding and/or following row. It is further contemplated that the stent **100** need not be formed from a single strand **120** but rather may include two or more strands knitted together. In some instances, a loop may be generally aligned with, or suspended from, a loop of the preceding row in a direction generally parallel to a longitudinal axis of the stent **100** (for example, circumferentially aligned along a length of the stent **100**). As can be seen, the loop **146b** in one row **136** is suspended from the loop **144b** in the row **134** above it. Thus, the loops may form axially extending columns or wales **150a-e**, although this is not required.

[0087] To form the stent **100**, an end region **154** of the strand **120** is passed over an intermediate rung portion **152b** of a preceding row **136**, as shown at arrow **156**. The end region **154** of the strand **120** may then be wrapped behind the loop **146c** in a direction opposite to the general direction **160** of the overall knit. The end region **154** of the strand **120** may then be passed over a rung portion **152a** on opposing side of the loop **146c** (relative to the rung portion **152b**) before being crossed over itself to complete the loop. The reverse configuration is also contemplated in which the loop passes behind the rung portions **152b**, **152a** and over the loop **146c**. The loops **140a-c**, **142a-c**,

144a-c, 146a-c, 148d-c may generally take the form of a twisted knit stitch where each individual loop is twisted. It is contemplated that the twisted nature of the loops may create ridges in the outer surface of the stent **100**. These ridges may help secure the stent **100** within the body lumen.

[0088] FIG. 5 is a perspective view of a mandrel **200** for forming a stent having anti-migration features, such as the stent **10** described herein. Some illustrative mandrels are described in commonly assigned U.S. Patent Publication Number 2019/0029850, entitled ADJUSTABLE MANDREL FOR FORMING STENT WITH ANTI-MIGRATION FEATURES, the disclosure of which is hereby incorporated by reference. As described above, in some instances, the stent may additionally include a tapered outer profile region with one or more flared end regions as well as anti-migration features. In some cases, the stent may be considered as having an hourglass profile, for example. However, in other instances, the stent may have a generally constant outer diameter with one or more anti-migration features extending radially outward therefrom.

[0089] As can be seen, the mandrel **200** may include a mandrel body **202**, a mandrel cap **204**, an actuation element **206** and a plurality of anti-migration feature forming pins **208**. FIG. 6 is an exploded perspective view of the mandrel **200**, with the anti-migration feature forming pins excluded for clarity. In some cases, the mandrel cap **204** may be releasably secured to the mandrel body **202** via a bolt **218**, bayonet coupling, or other securement mechanism. In some instances, the mandrel cap **204** may be removable from the mandrel body **202** in order to facilitate removal of a stent from the mandrel **200**. In other embodiments, the mandrel body **202** and the mandrel cap **204** may be formed as a unitary or monolithic structure, particularly if the mandrel cap **204** has an outer diameter roughly the same as an outer diameter of the mandrel body **202**. In some instances, the mandrel body **202** may include a cylindrical portion having an outer diameter and the mandrel cap **204** may have an outer diameter greater than the cylindrical portion of the mandrel body **202**.

[0090] The mandrel body **202**, shown in FIGS. 5 and 6 may include a first stent shaping segment **210** and a second stent shaping segment **212**. In some cases, the mandrel cap **204** may be a third stent shaping segment **211**. The first stent shaping segment **210** may be a cylindrical portion of the mandrel body **202** having a first diameter the second stent shaping segment **212** may be a cylindrical portion of the mandrel body having a second diameter, and the third stent shaping segment **211** may be a cylindrical portion of the mandrel cap having a third diameter. In some cases, the first stent shaping segment **210**, the second stent shaping segment **212**, and/or the third stent shaping segment **211** may have a non-cylindrical profile. For example, the first stent shaping segment **210**, the second stent shaping segment **212**, and/or the third stent shaping segment **211** may instead have a polygonal cross-sectional profile such as an octagonal cross-sectional profile. This is just an example. The first, second and third diameters may be selected based on the desired shape of the stent in the expanded configuration. For example, the mandrel **200** illustrated in FIG. 5 may produce a stent having a generally hourglass shape. The first diameter may be different than the second diameter and/or the third diameter. For example, the first diameter may be greater than the second diameter. However, this is not required. In some cases, the first diameter and the second and/or third diameter may be the same or substantially the same. When the first and second diameter are different, a tapered segment **214** extends between the first stent shaping segment **210** and the second stent shaping segment **212** and defines a tapered surface **216** extending from a cylindrical outer surface of the first stent shaping segment **210** to a cylindrical outer surface of the second stent shaping segment **212**. A similar tapered segment may extend between the second diameter and the third diameter. The tapered segment **214** includes a plurality of apertures **220** that extend through the circumferential wall of the tapered segment from the tapered surface **216** to an internal bore **228** (see, for example, FIGS. 7 and 8) extending axially within the mandrel body **202** in order to accommodate the anti-migration feature forming pins **208**. It will be appreciated that an angle of the tapered surface **216**, relative to the first stent shaping segment **210** and/or the second stent shaping segment **212**, may influence the relative angle at which the anti-migration feature forming pins **208** extend outwardly from the tapered surface **216**. In some cases, particularly if the

first stent shaping segment **210**, the second stent shaping segment **212**, and/or the third stent shaping segment **211** have a similar or identical outer diameter, the tapered surface **216** may itself not be tapered, but may instead have a constant outer diameter.

[0091] In some instances, at least some of the plurality of apertures **220** may have a major dimension that is orthogonal to the tapered surface **216**. In some cases, at least some of the plurality of apertures **220** may have a major dimension that extends at an acute angle relative to the tapered surface **216**. It will be appreciated that in some cases, some of the plurality of apertures **220** may extend at different angles relative to the tapered surface **216**. As shown, the plurality of apertures **220** may be considered as being radially aligned in a ring that extends around the tapered segment **214**. In some cases, it will be appreciated that some of the plurality of apertures **220** may be axially displaced relative to others of the plurality of apertures **220**. In other words, some of the plurality of apertures **220** may form a first ring around the tapered segment **214** while others of the plurality of apertures **220** may form a second ring around that tapered segment **214** that is axially displaced from the first ring around the tapered segment **214**.

[0092] In some cases, at least some of the plurality of apertures **220** may extend linearly through the tapered segment **214** such that each corresponding pin **208** extends through the aperture **220** orthogonally to the tapered surface **216**. In some cases, at least some of the apertures **220** may have a curved or helical shape, such that as the corresponding pin **208**, which may have a complementary curved or helical shape, is extended out of the aperture **220**, the pin **208** may rotate, and thus a distal end of the pin **208** may move radially as well as axially.

[0093] The actuation element **206** may be configured to extend into the bore **228** of the mandrel body **202** from one end of the mandrel body **202** (e.g., the end of the mandrel body opposite to the mandrel cap **204**) to selectively engage and actuate the pins **208** within the apertures **220**. For example, the actuation element **206**, shown FIG. 6, includes a tapered end **222** that may be configured to engage the anti-migration feature forming pins **208**, as well as a threaded body **224** that is configured to threadedly engage a threaded aperture extending within the first stent shaping segment **210** of the mandrel body **202**. In some instances, the tapered end **222** may be conically, frustoconically, convexly, or concavely tapered. The actuation element **206** may be considered as including a handle **226** that may be used by an individual or a machine to rotate the actuation element **206** and thus advance the actuation element **206** into the bore of the mandrel body **202** by rotating in a first direction or withdraw the actuation element **206** from the bore of the mandrel body **202** by rotating in a second, opposite direction. Thus, the threaded body **224** may threadably engage a threaded region of the bore of the mandrel body **202** to threadably advance the actuation element **206** into the bore (e.g., toward the mandrel cap **204**) by rotating the actuation element **206** in a first rotational direction and withdraw the actuation element **206** from the bore (e.g., away from the mandrel cap **204**) by rotating the actuation element **206** in a second, opposite rotational direction. This may be demonstrated, for example, with respect to FIGS. 7 and 8, which are cross-sectional views showing the actuation element **206** fully extended into the bore **228** of the mandrel body **202** (FIG. 7) or partially extended (FIG. 8) taken along line 7-7 of FIG. 5. In other cases, it is contemplated that rather than the actuation element **206** itself including a threaded region, a threaded fastener may be configured to engage a threaded bore **228** of the mandrel body **202** to actuate the actuation element **206** relative to the mandrel body **202**.

[0094] FIG. 7 shows the actuation element **206** fully extended into the bore **228** of the mandrel body **202** with the threaded body **224** threadably engaged with the threaded region of the bore of the mandrel body **202**. In particular, the bore of the mandrel body **202** includes a first threaded region **230** extending into the first stent shaping segment **210** of the mandrel body **202** from a first end of the mandrel body **202** that is configured, in diameter, depth and thread pitch, to threadably engage the threaded body **224** of the actuation element **206**. In some instances, as illustrated, the mandrel body **202** also includes a second threaded bore or region **232** extending into the second stent shaping segment **212** of the mandrel body **202** from the second, opposite end of the mandrel

body that is configured, in diameter depth and thread pitch, to threadably engage threads on the threaded fastener (e.g., bolt or screw) **218** in order to releasably secure the mandrel cap **204** in position relative to the mandrel body **202** at the second end of the mandrel body **202**. In some cases, it is contemplated that rather than utilizing a separate threaded fastener **218**, that the mandrel cap **204** itself may include a threaded protuberance that is configured to engage the second threaded bore **232**. Alternatively, it is also contemplated that the second end of the mandrel body **202** may include a threaded protuberance, and the mandrel cap **204** may include a threaded bore or aperture to engage the threaded protuberance of the mandrel body **202**, or a through hole for passing the threaded protuberance through to be threadably engaged with a mating threaded fastener (e.g., nut) on an opposite side of the mandrel cap **204**. In either event, the mandrel cap **204** may be secured to or removed from the mandrel body **202**, particularly for aid in removing a formed stent from the mandrel **200**. In some cases, the mandrel cap **204** may be permanently secured to the mandrel body **202**, particularly in cases where the mandrel **200** has a profile in which an outer diameter of each successive stent shaping segment is equal to or less than an outer diameter of a preceding stent shaping segment and a formed stent may simply be slid off the mandrel **200** without removing the mandrel cap **204**. In some cases, the mandrel body **202** may include a locating or centering aperture **234** that is configured to accommodate a locating or centering feature **236** extending from the mandrel cap **204**, but this is not required in all cases. In some cases, rather than using the fastener **218** to secure the mandrel cap **204** to the mandrel body **202**, the locating or centering feature **236** may itself threadably engage the locating or centering aperture **234**.

[0095] As shown in FIG. 7, the actuation element **206** is fully extended into the first threaded region **230** of the bore of the mandrel body **202**. As a result, the anti-migration feature forming pins **208** can be seen as being extended radially outwardly through the corresponding apertures **220**. In some cases, depending on the particular dimensions of the various components forming the mandrel **200**, the anti-migration feature forming pins **208** may be considered as being extended radially outwardly as far as they can go before the actuation element **206** is fully extended into the first threaded region **230** of the bore of the mandrel body **202**. A base **238** of each pin **208** may be seen as engaging the tapered end **222** of the actuation element **206**. This can be contrasted with FIG. 8, in which the actuation element **206** is only partially extended into the first threaded region **230** of the bore of the mandrel body **202**. Accordingly, while the base **238** of each pin **208** (only 2 pins are shown for clarity) is still engaged with the tapered end **222** of the actuation element **206**, it can be seen that the pins **208** do not extend radially outwardly through the corresponding apertures **220** as far as the pins **208** extend in FIG. 7. In some cases, as shown in FIGS. 7 and 8, the base **238** of each pin **208** may be larger in at least one dimension than a diameter of the corresponding aperture **220**. Thus, the extent that the pins **208** can be extended radially outward through the apertures **220** may be limited when the base **238** of the pin **208** abuts a peripheral edge of the aperture **220**. As a result, the pins **208** are retained within the apertures **220** and won't fall out. The pins **208** can, in some instances, be removed completely by withdrawing the actuation element **206** from the bore of the mandrel body **202**, permitting the pins **208** to move radially inward of the apertures **220** and then fall into the bore **228** of the mandrel body **202**.

[0096] FIG. 9 is a perspective view of one example of an anti-migration feature forming pin **208**. In some cases, the pin **208** may include a pin body **240** extending between the base **238** (which may have an enlarged cross-section relative to the pin body **240**) and a pin end **242** opposite the base **238**. As noted, the base **238** may be larger in diameter than the pin body **240**, but this is not required in all cases. It is contemplated that the number of anti-migration forming pins **208** and the size, shape, and/or orientation (e.g., angle) thereof may be varied to achieve an anti-migration features having a desired shape and orientation. In some cases, the pin end **242** may be curved to facilitate a portion of a wire of a stent to be formed in a curved shape. In some cases, the curved shape may be a simple curve. In some instances, the curved shape may be a compound curve, such

as an undulating or wave-like shape. In some instances, the pin end **242** may include a recessed slot **244** that may be configured to accommodate a wire or wires of the stent being shaped on the mandrel **200**. In some cases, the recessed slot **244** may itself have a simple or compound curve shape to instill a corresponding simple or compound curve shape to a stent wire extending through the recessed slot **244**. For example, in some embodiments the recessed slot **244** may be a curved slot **244** providing a wire placed therein with a curved region. In some cases the recessed slot **244** may include two converging portions converging at a point at the pin end **242** to provide a wire with a sharp bend for an anti-migration feature. In some cases, the stent being formed is a knitted stent, and a constant diameter knitted stent blank, such as stent **10** or **100**, may be stretched over the mandrel **200**, with a particular wire of the knitted stent blank disposed within the recessed curved slot **244** in order to form an anti-migration feature extending radially outward from a knitted tubular wall of the stent. In other cases, the stent may be knitted directly over the mandrel **200**. [0097] While the pin end **242** is illustrated as a curved profile and being no larger in dimension than the pin body **240**, in some cases it is contemplated that the pin end **242** may extend laterally beyond the pin body **240** and form an arcuate surface. In some cases, for example, the arcuate surface of each of the pin ends **242** may align end to end, and essentially form a raised ring extending around the mandrel **200**. The individual arcuate surfaces of each of the pin ends **242** maybe driven outward by extending the actuation element **206** into the mandrel body **202** by rotating the actuation element **206** in a first rotational direction in order to form a raised ring anti-migration feature in the stent. Rotating the actuation element **206** in a second, opposing rotational direction, allows the pins **208** to retract, and allow removal of the stent from the mandrel **200**. [0098] FIG. **10** shows a portion of a knitted stent **300** disposed on the mandrel **200**, while FIG. **11A** shows the knitted stent **300** removed from the mandrel **200**. The stent **300** may be similar in form and function to the stents **10**, **100** described above. The stent **300** may have a first, or proximal end **320**, a second, or distal end **322**, and an intermediate region **324** disposed between the first end **320** and the second end **322**. Generally, the stent **300** may be formed from a single knitted strand or filament **302**, as described above with respect to stents **10**, **100**. The stent **300** includes a plurality of loops **304** each with an overlapping base portion **305** and a plurality of interconnecting rung sections **306**. As shown in FIG. **10**, one of the rung sections **306** of the knitted stent **300** may extend radially outward from the knitted tubular wall of the stent **300** and along the recessed slot **244** of the pin **208** to form one or more of the anti-migration features **308** of the stent **300**. Thus, the outwardly extending rung section **306** forming the anti-migration feature **308** may extend radially outward relative to rung sections **306** longitudinally and/or circumferentially adjacent thereto. In some cases, a knitted stent such as the knitted stent **300** may be formed by first knitting a constant diameter stent blank (not illustrated), then stretching the constant diameter stent blank over the mandrel **200** prior to a shaping process and/or an annealing process. It can be seen that the knitted stent **300** has a first enlarged diameter portion **310** proximate a first end of the knitted stent **300** that corresponds to the first stent shaping segment **210**, a second enlarged diameter portion **314** proximate a second end of the knitted stent **300** that corresponds to the third stent shaping segment **211**, and a (relatively) reduced diameter portion **312** (e.g., a cylindrical body region intermediate the first and second enlarged diameter portions **310**, **314**) that corresponds to the second stent shaping segment **212**.

[0099] The knitted stent **300** includes anti-migration features **308** that correspond to the pins **208** which are arranged circumferentially around the knitted stent **300** at a transition region between the first enlarged diameter portion **310** and the reduced diameter portion **312**. However, it is contemplated that the anti-migration features **308** may be arranged at a different location along the length of the knitted stent **300**, if desired. The pins **208** may be actuated radially outward with the wires disposed in the recessed slots **244** after the knitted stent blank has been placed on the mandrel **200** to cause the portions of the wires engaged with the pins **208** to be urged radially outward from the knitted tubular wall of the stent to form the anti-migration features **308**. The anti-migration

features **308** may extend in a generally distal direction. However, this is not required. It is contemplated that the general orientation of the anti-migration features **308** can be changed as desired by changing an orientation or shape of the pins **208**. For example, FIG. **11B** illustrates a perspective view of an illustrative stent **300b** having an alternative configuration for the anti-migration features **308b**. In the illustrative embodiment of FIG. **11B**, the anti-migration features **308b** are oriented in a generally proximal direction. FIG. **11C** illustrates a perspective view an illustrative stent **300c** having yet another alternative configuration for the anti-migration features **308c**. In the illustrative embodiment of FIG. **11C**, the anti-migration features **308c** initially extend in a generally distal direction and are then turned back on themselves such that an apex of the anti-migration features **308c** is pointed in a more proximal direction. Other orientations or combinations of orientations may be used, as desired.

[0100] FIG. **12A** is an end view of the knitted stent **300**, showing the anti-migration features **308** extending radially outward from the knitted tubular wall of the knitted stent **300**. As illustrated, each of the anti-migration features **308** are loops of the filament(s) or wire(s) forming the knitted stent **300** extending between adjacent overlapping base portions **305**, each loop being roughly equal in shape and dimension. As described above, overlapping base portions **305** may be the location in which portions of the filament(s) or wire(s) cross or loop around another portion of the filament(s) or wire(s). In other cases, some of the anti-migration features **308** may vary in shape and/or dimension, or may not be equally spaced, for example. While the anti-migration features **308** are shown as being curved, in some cases the anti-migration features **308** may be pointed, or include a flattened region, for example. FIG. **12B**, for example, shows a knitted stent **300d** that includes a number of anti-migration features **308d**. The anti-migration features **308d** each extend between adjacent overlapping base portions **305**, and are each roughly equal in shape and dimension. However, by comparing FIG. **12B** with FIG. **12A**, it can be seen that the anti-migration features **308** shown in FIG. **12A** extend radially outward further than the anti-migration features **308d** shown in FIG. **12B**. The anti-migration features **308d** may be formed, for example, by using anti-migration feature forming pins **208** that are shorter in length, or by not advancing the actuation element **206** as far into the mandrel body **202**, thus not advancing the pins **208** radially outward as far from the surface of the tapered segment **214** of the mandrel body **202**. The anti-migration features **308d** may be pointed, for example, or have other shapes as well.

[0101] It will be appreciated that the relative dimensions of the anti-migration features **308** and the anti-migration features **308d** may be a function of the ultimate end-use of the knitted stent **300** (or **300d**). Relatively larger anti-migration features **308**, **308d** may be useful in situations where the knitted stent **300** (or **300d**) will be placed in anatomical locations where the knitted stent **300** (or **300d**) may be subjected to relatively stronger migration forces and/or anatomical locations where the dimensions of the patient's anatomy are more variable. Relatively smaller anti-migration features **308**, **308d** may be useful in situations where the knitted stent **300** (or **300d**) may be subjected to relatively weaker migration forces and/or anatomical locations where the dimensions of the patient's anatomy are less variable. In some cases, the overall dimensions of the knitted stent **300** (or **300d**) may play a part as well. In some cases, for example, a larger diameter knitted stent **300** (or **300d**) may have relatively larger anti-migration features **308**, **308d** while a smaller diameter knitted stent **300** (or **300d**) may have relatively smaller anti-migration features **308**, **308d**.

[0102] FIG. **12C** shows a knitted stent **300e** that includes a number of anti-migration features **308c**. In contrast to the knitted stents **300** and **300d** shown in FIGS. **12A** and **12B**, the anti-migration features **308e** do not extend between each adjacent overlapping base portion **305** about the periphery of the knitted stent **308b**. Each of the anti-migration features **308c** extends between adjacent overlapping base portion **305**, although some overlapping base portion **305** are not attached to an anti-migration feature **308e**. The anti-migration features **308e** illustrated in FIG. **12C** may have an alternating pattern (e.g., alternating between an anti-migration features **308e** and a rung section **306**). Other patterns or asymmetric arrangements can be used as desired. As

illustrated, each of the anti-migration features **308c** are roughly equal in shape and dimension. The anti-migration features **308c** may be formed, for example, by only placing anti-migration feature forming pins **208** into some of the apertures **220**. In some cases, it is contemplated that some of the anti-migration features **308c** may be smaller or larger in dimension, and/or may vary in shape, relative to others of the anti-migration features **308c**.

[0103] FIG. **12D** shows a knitted stent **300f** that includes a number of anti-migration features **308f** and a number of anti-migration features **308g**, each extending between adjacent overlapping base portions **305**. It will be appreciated that as illustrated, each of the anti-migration features **308f** are roughly equal in shape and dimension, and each of the anti-migration features **308g** are roughly equal in shape and dimension, albeit not extending radially outward as far as the anti-migration features **308g**. The anti-migration features **308f** and **308g** may be formed, for example, by using a longer length pin **208** to form each of the anti-migration features **308f** and a shorter length pin **208** to form each of the anti-migration features **308g**. It will be appreciated that the particular anti-migration features **308**, **308b**, **308c**, **308d**, **308c**, **308f**, and **308g** shown in FIGS. **11A-11C** and **12A-12D** are merely illustrative, and may be mixed or matched in any desired pattern.

[0104] FIG. **13** is a partial side view of another knitted stent **400** having one or more anti-migration features **408**. The stent **400** may be similar in form and function to any of the stents described herein. For example, the stent **400** may have a generally constant diameter from a proximal end to a distal end similar to the stents **10**, **100** described above. Alternatively, the stent **400** may include one or more regions having an increased diameter similar to the stent **300** described above. The stent **400** may be formed from a single knitted strand or filament **402**. The strand **402** may be manipulated (e.g., knitted) into a plurality of rows each having a plurality of interconnected or intermeshing loops **404a-f** (collectively, **404**) each with an overlapping base portion **405**. The stent **400** may include as many rows as required to form a stent **400** having the desired length. As described above, the loops **404** may be loosely knit and include interconnecting intermediate rung portions **406** between the loops **404**. It is further contemplated that the stent **400** need not be formed from a single strand **402** but rather may include two or more strands knitted together. In some instances, a loop may be generally aligned with, or suspended from, a loop of the preceding row in a direction generally parallel to a longitudinal axis of the stent **400** (for example, circumferentially aligned along a length of the stent **100**). Thus, the loops **404** may form axially extending columns or wales, although this is not required.

[0105] As described above, some (or all) of the intermediate rung portions **406** may be formed into anti-migration features **408**. For example, one or more rung portions **406** may be biased to radially extend from the main body of the stent **400** using a mandrel, such as the mandrel **200** described herein. It is contemplated that while the anti-migration features **408** are being formed, a point bend **410a**, **410b** (collectively, **410**) may be formed in one or more of the anti-migration features **408**. The point bends **410** may be a pinch point or a region where the anti-migration features **408** undergo an abrupt change in direction as opposed to a smooth curve. The point bends **410** may be formed near or at the intersection of the anti-migration features **408** with the adjacent loop **404a**, **404b** when the stent is in a radially expanded configuration. The portion of the strand **402** forming the anti-migration features **408** and including the point bends **410** may pass under the adjacent loop **404a**, **404b** to tie the anti-migration feature **408** back into the stent **400**. It is contemplated that the angle of the point bends **410** may be selected to provide a desired angle or orientation of the anti-migration features **408**. The point bends **410** may be formed using anti-migration feature forming pins **208**. For example, the pins **208** may include a recess along a lower edge thereof to form the point bend. The stent **400** may be annealed or heat treated after formation of the anti-migration features **408** and/or point bends **410**.

[0106] When the stent **400** is deployed, the anti-migration features **408** radially protrude from the stent **400**, as shown in FIG. **14**, which illustrates a side view of region A of the stent **400** of FIG. **13**. As can be seen in FIG. **14**, the point bends **410** may be positioned underneath or adjacent to the

loops **404a**, **404b** in the radially expanded configuration. While the anti-migration feature **408** is illustrated as extending generally orthogonal to longitudinal axis of the stent **400**, the anti-migration features **408** may extend at an angle between about 0° and 180° relative to the longitudinal axis of the stent **400**. The point bends **410** may be formed between a first portion or arm **414a**, **414b** (collectively, **414**) of the filament and a second portion or arm **416a**, **416b** (collectively, **416**). The first arms **414** may be a portion of the filament exiting the adjacent loops **404** while the second arms **416** may together form the anti-migration features **408**. The point bends **410** may be formed such that the second portion **416** extends at an angle **412a**, **412b** (collectively, **412**) relative to the first portion **414**. The angle **412** may determine the angle and/or direction the anti-migration features **408** are oriented. For example, in the illustrated example, the angle **412** is around 90° and thus the anti-migration features **408** extend at an angle of about 90° relative to the longitudinal axis of the stent **400**. If the point bends **410** are formed such that the angle **412** is obtuse, the anti-migration features **408** may be pointed in a generally distal direction **418**. For example, in some instances the angle **412** may be greater than 90° but less than 155° , greater than 90° but less than 145° , greater than 90° but less than 135° , greater than 90° but less than 120° , greater than 100° but less than 155° , greater than 100° but less than 145° , greater than 100° but less than 135° , or greater than 100° but less than 120° . If the point bends **410** are formed such that the angle **412** is acute, the anti-migration features **408** may be pointed in a generally proximal direction **420**. For example, in some instances the angle **412** may be less than 90° but greater than 25° , less than 90° but greater than 35° , less than 90° but greater than 45° , less than 90° but greater than 55° , less than 80° but greater than 25° , less than 80° but greater than 35° , less than 80° but greater than 45° , or less than 80° but greater than 55° .

[0107] It is contemplated the angle **410** may be determined by a radius of curvature of the point bends **410**. A smaller radius of curvature may result in a sharper angle **412** while a larger radius of curvature may result in a less sharp angle **412**. In some instances, the radius of curvature of the angle **412** may be less than four times the diameter of the strand or filament, may be less than three times the diameter of the strand or filament, may be less than two times the diameter of the strand or filament, or the radius of curvature of the angle **412** may be less than or equal to the diameter of the strand or filament. In some embodiments, the point bends **410** may be formed to have similar angles **412** (and/or radii of curvature). In other embodiments, some anti-migration features **408** may be formed with a first angle **412** and other anti-migration features **408** may be formed with a second angle, different from the first **412**. This may allow the anti-migration features **408** to have differing radial profiles (as shown in FIG. 12D). It is contemplated that any number of angles **412** may be combined in varying patterns to achieve the desired effect.

[0108] Referring additionally to FIG. 15, which illustrates the stent **400** of FIG. 14 under an applied force **450**, when a removal force is applied to the stent **400**, for repositioning or removal of the stent **400**, the stent **400** may begin to elongate (in a similar manner to that show in FIG. 3). As the stent **400** elongates, the point bends **410** are pulled through the adjacent loops **404a**, **404b**. As the point bends **410** are passed through the loops **404**, the anti-migration feature **408** is deflected radially inward, as shown at arrow **452**, to lie flat along the body of the stent **400**. Thus, a portion of the anti-migration feature **408** (portions of the strand or filament forming the anti-migration feature **408** between adjacent point bends **410**, may also be pulled through the adjacent loops **404a**, **404b**. In some instances, as the stent **400** elongates, the angle **412** at the point bend **410** may increase as the anti-migration feature **408** is deflected radially inward toward the outer diameter of the body of the stent **400**. It is contemplated that each anti-migration feature **408** that is positioned at a similar longitudinal location along the stent **10** may deflect down at a same point of elongation of the stent **10**. These features may reduce the force required to move or remove the stent **400** from the body or tissue. It is contemplated that the direction of the applied force may vary based on the direction in which the anti-migration features **408** are oriented and/or an angle of the point bends **410**. For example, distally oriented anti-migration features **408** may be more readily disengaged

from the tissue with a proximally applied force while proximally oriented anti-migration features **408** may be more readily disengaged from the tissue with a distally applied force.

[0109] To form a knitted stent, such as the stents **10**, **100**, **300**, **300b-f**, **400** described herein having a non-uniform profile and one or more anti-migration features, a constant diameter knitted stent blank (as shown in FIG. **4**) may be positioned over a mandrel (such as mandrel **200**) having a tapered outer surface and one or more anti-migration feature forming elements. It should be noted that a knitted stent having a generally uniform diameter and one or more anti-migration features can be formed by place a constant diameter knitted stent blank over a generally uniform mandrel with one or more anti-migration feature forming elements. In other embodiments, a knitted stent having a non-uniform or a uniform profile may be knitted directly over the mandrel having one or more anti-migration feature forming elements. In some cases, disposing a constant diameter knitted stent blank in position over a mandrel includes stretching the constant diameter knitted stent blank over the mandrel and allowing the constant diameter knitted stent blank to conform to the varied diameter outer surface of the mandrel (e.g., conforming to the various constant diameter regions and/or tapered diameter regions of the mandrel).

[0110] The one or more anti-migration feature forming elements (such as but not limited to the pins **208**) may be engaged by a portion of the stent in order to provide a desired shape prior to annealing. In some cases, the one or more anti-migration feature forming elements are pins that are configured to be driven in a radially outward direction relative to the outer surface of the mandrel. Engaging the one or more anti-migration feature forming elements with the stent may include driving the pins in radially outward direction relative to the mandrel to urge the wire(s) or filament(s) engaged with the end of each of the pins in a radially outward direction relative to the knitted tubular structure of the stent. The mandrel and the stent thereon, with the anti-migration features formed, may then be subjected to an annealing or shape setting process. After the annealing or shape setting process, the one or more anti-migration feature forming elements may be disengaged in order to remove the shaped stent from the mandrel. In some cases, disengaging the one or more anti-migration feature forming elements comprises permitting the pins to move in an inward direction relative to the mandrel.

[0111] In some cases, it may be desirable to form knitted stent with one or more anti-migration features from a metallic component and a non-metallic or even biodegradable component. In such a case, the metallic component and the non-metallic component may individually be shaped, and then combined to form a stent. In some cases, each of the metallic component and the non-metallic or even biodegradable component may each include anti-migration features, where the anti-migration features of the non-metallic or even biodegradable component complement the anti-migration features of the metallic component. In cases where the non-metallic component is biodegradable, the biodegradable anti-migration features may provide additional resistance to migration upon initial implantation of the stent, but dissolve away over time.

[0112] To form a two (or more) component stent, in some cases, a constant diameter metallic knitted stent blank may be positioned over a mandrel having a tapered outer surface and one or more anti-migration feature forming elements. The mandrel may be the mandrel **200**, for example. However, it is not required for the mandrel to have tapered surfaces or varying diameters. In some cases, disposing a constant diameter metallic knitted stent blank in position over a mandrel includes stretching the constant diameter metallic knitted stent blank over the mandrel and allowing the constant diameter metallic knitted stent blank to conform to the varied diameter outer surface of the mandrel (e.g., conforming to the various constant diameter regions and/or tapered diameter regions of the mandrel).

[0113] The one or more anti-migration feature forming elements (such as but not limited to the pins **208**) may be engaged with a portion of the metallic knitted stent in order to provide a desired shape prior to annealing. In some cases, the one or more anti-migration feature forming elements are pins that are configured to be driven in a radially outward direction relative to the outer surface of the

mandrel. For example, to engage a portion of the stent the pins are driven in radially outward direction relative to the mandrel to urge the wire(s) or filament(s) engaged with the end of each of the pins in a radially outward direction relative to the knitted tubular structure of the stent. The mandrel and the stent thereon, with the anti-migration features formed, may then be subjected to an annealing or shape setting process. After the annealing or shape setting process, the one or more anti-migration feature forming elements may be disengaged in order to remove the shaped stent from the mandrel. In some cases, disengaging the one or more anti-migration feature forming elements comprises permitting the pins to move in an inward direction relative to the mandrel.

[0114] In some cases, once the shaped metallic stent has been removed from the mandrel, a constant diameter biodegradable knitted stent blank may be positioned over a mandrel having a tapered outer surface and one or more anti-migration feature forming elements. In some cases, disposing a constant diameter biodegradable knitted stent blank in position over a mandrel includes stretching the constant diameter biodegradable knitted stent blank over the mandrel and allowing the constant diameter biodegradable knitted stent blank to conform to the varied diameter outer surface of the mandrel. The one or more anti-migration feature forming elements may be engaged in order to provide a desired shape prior to annealing.

[0115] In some cases, the annealing process for the biodegradable knitted stent blank may involve lower temperatures than that used for the metallic knitted stent blank. The mandrel and the stent thereon, with the anti-migration features formed, may then be subjected to an annealing or shape setting process. After the annealing or shape setting process, the one or more anti-migration feature forming elements may be disengaged in order to remove the shaped biodegradable stent from the mandrel. In some cases, disengaging the one or more anti-migration feature forming elements comprises permitting the pins to move in an inward direction relative to the mandrel. In some cases, while not illustrated, the shaped biodegradable stent may be disposed about or within the shaped metallic stent.

[0116] FIG. **16** is a side view of an illustrative delivery system **500** for delivering a stent, such as the stents **10**, **100**, **300**, **300b-f**, **400** described herein, to a target region. The delivery system **500** may include an outer or exterior elongate shaft or tubular member **502** and an inner elongate shaft or tubular member **504**. The inner tubular member **504** may be slidably disposed within a lumen of the outer tubular member **502**. The outer tubular member **502** may extend proximally from a distal end region **506** to a proximal end region **508** configured to remain outside of a patient's body. A first hub or handle **510** may be coupled to the proximal end region **508** of the outer tubular member **502**. The inner tubular member **504** may extend proximally from a distal end region **512** to a proximal end region **514** configured to remain outside of a patient's body. A second hub or handle **516** may be coupled to the proximal end region **514** of the inner tubular member **504**. In some instances, the distal end region **506** of the outer tubular member **502** may be configured to be atraumatic.

[0117] The outer tubular member **502** may include a lumen **518** extending from the distal end region **506** to the proximal end region **508**. The lumen **518** may also extend through the first handle **510**. The lumen **518** of the outer shaft **502** and the first handle **510** may be configured to slidably receive the inner shaft **504**. The inner tubular member **504** may include a lumen **520** extending from the distal end region **512** to the proximal end region **514**. The lumen **520** of the inner tubular shaft **504** may also extend through the second handle **516**. The lumen **520** of the inner shaft **504** may be configured to receive a guidewire **522**, as desired.

[0118] The stent **10** may be disposed around a portion of the inner tubular member **504** at or adjacent to the distal end region **512** thereof. When the stent **10** is disposed over the inner tubular member **504**, in a collapsed and elongated delivery configuration, the stent **10** may be restrained in a collapsed reduced diameter or delivery configuration by the outer tubular member **502** surrounding the stent **10**. In the collapsed configuration, the stent **10** may have a smaller diameter and a longer length than the expanded deployed configuration. The distal end region **506** of the

outer tubular member **502** may be positioned such that the outer tubular member **502** surrounds and covers the length of the stent **10** during delivery. The outer tubular member **502** may have sufficient hoop strength to retain the stent **10** in its reduced diameter state.

[0119] FIG. **17** illustrates a side view of the delivery system **500** with the stent **10** in a partially deployed configuration. The delivery system **500** may be advanced through the gastrointestinal tract (or other body lumen), as desired. The delivery system **500** may be advanced with or without the use of a guidewire **522**. Once the stent **10** is positioned adjacent to the target region, the restraining forces maintaining the stent **10** in the radially collapsed configuration may be removed to deploy the stent **10**.

[0120] The stent **10** may be released by actuating the first handle **510** proximally relative to the second handle **516**, e.g., by pulling the first handle **510** proximally **524** while maintaining the second handle **516** in a fixed position. Thus, the outer tubular shaft **502** may be retracted proximally relative to the inner tubular shaft **504**. In other words, the outer tubular shaft **502** may be proximally retracted while the inner tubular shaft **504** is held stationary. As shown in FIG. **17**, as the outer tubular shaft **502** is retracted proximally **524** to uncover the stent **10**, the biasing force is removed from the exterior of the stent **10** and the stent **10** assumes its radially expanded, unbiased, deployed configuration. Once the outer tubular member **502** no longer covers the proximal end **14** of the stent **10**, the stent **10** may assume its fully deployed configuration, as shown in FIG. **1**. The delivery system **500** may then be removed from the body lumen.

[0121] The stents, delivery systems, and the various components thereof, may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material. Some examples of suitable metals and metal alloys include stainless steel, such as 304V, 304L, and 316LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic Nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys, nickel-copper alloys, nickel-cobalt-chromium-molybdenum alloys, nickel-molybdenum alloys, other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys; platinum enriched stainless steel; titanium; combinations thereof; and the like; or any other suitable material.

[0122] Some examples of suitable polymers for the stents or delivery systems may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly (alkylene ether) phthalate and/or other polyester elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), MARLEX® high-density polyethylene, MARLEX® low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro (propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly (styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof,

polymer/metal composites, and the like.

[0123] In at least some embodiments, portions or all of the stents or delivery systems may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are generally understood to be materials which are opaque to RF energy in the wavelength range spanning x-ray to gamma-ray (at thicknesses of $<0.005''$). These materials are capable of producing a relatively dark image on a fluoroscopy screen relative to the light image that non-radiopaque materials such as tissue produce. This relatively bright image aids the user of the stents or delivery systems in determining its location. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the stents or delivery systems to achieve the same result.

[0124] It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A stent, the stent comprising: an elongated tubular member comprising at least one knitted filament having a plurality of twisted knit stitches with intermediate rung portions extending between adjacent twisted knit stitches, the elongated tubular member configured to move between a collapsed configuration and an expanded configuration; and one or more anti-migration features formed in one or more of the intermediate rung portions, wherein the one or more anti-migration features each includes a first point bend adjacent a loop of a first twisted knit stitch and a second point bend adjacent a loop of a second twisted knit stitch; wherein the first point bend is formed between a first arm of the knitted filament and a second arm of the knitted filament, and the second point bend is formed between the second arm of the knitted filament and a third arm of the knitted filament; wherein when the elongated tubular member is in the expanded configuration, the one or more anti-migration features extend at an angle relative to a longitudinal axis of the stent.
2. The stent of claim 1, wherein the second arm of the knitted filament forms the one or more anti-migration features.
3. The stent of any one of claim 1, wherein the first and second point bends are formed such that the second arm extends at a point bend angle relative to both the first arm of the knitted filament and the third arm of the knitted filament.
4. The stent of claim 3, wherein the point bend angle determines the angle that the one or more anti-migration features are oriented.
5. The stent of claim 4, wherein when the point bend angle is obtuse and the elongated tubular member is in the expanded configuration, the one or more anti-migration features are pointed in a distal direction.
6. The stent of claim 4, wherein when the point bend angle is acute and the elongated tubular member is in the expanded configuration, the one or more anti-migration features are pointed in a proximal direction.
7. The stent of claim 3, wherein the one or more anti-migration features each has a different point bend angle.
8. The stent of claim 1, wherein when a removal force is applied to the stent, the stent begins to elongate.
9. The stent of claim 8, wherein when the stent begins to elongate, the point bends subsume into one or more adjacent twisted knit stitches, and the one or more anti-migration features are deflected radially inward to lay flat along the elongated tubular member.

- 10.** The stent of claim 1, wherein when the elongated tubular member is in the collapsed configuration at least a portion of the one or more anti-migration features is subsumed into one or more adjacent twisted knit stitches.
- 11.** The stent of claim 1, wherein a length of the intermediate rung portions in the collapsed configuration is less than a length of the intermediate rung portions in the expanded configuration.
- 12.** The stent of claim 1, wherein at least some of the plurality of twisted knit stitches are suspended from a twisted knit stitch in a preceding row.
- 13.** The stent of claim 1, wherein a diameter of the elongated tubular member in the collapsed configuration is in the range of 60% to 80% less than a diameter of the elongated tubular member in the expanded configuration.
- 14.** The stent of claim 1, wherein when the stent is in the expanded configuration the one or more anti-migration features extend in a range of 1 to 4 millimeters radially beyond a base diameter of the elongated tubular member.
- 15.** The stent of claim 1, wherein the one or more anti-migration features include a non-metallic component attached to the knitted filament.
- 16.** The stent of claim 1, wherein a radius of curvature of the first point bend is less than four times a diameter of the at least one filament.
- 17.** The stent of claim 16, wherein a radius of curvature of the second point bend is less than four times a diameter of the at least one filament.
- 18.** The stent of claim 1, wherein a radius of curvature of the first point bend is less than two times a diameter of the at least one filament.
- 19.** The stent of claim 18, wherein a radius of curvature of the second point bend is less than two times a diameter of the at least one filament.
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